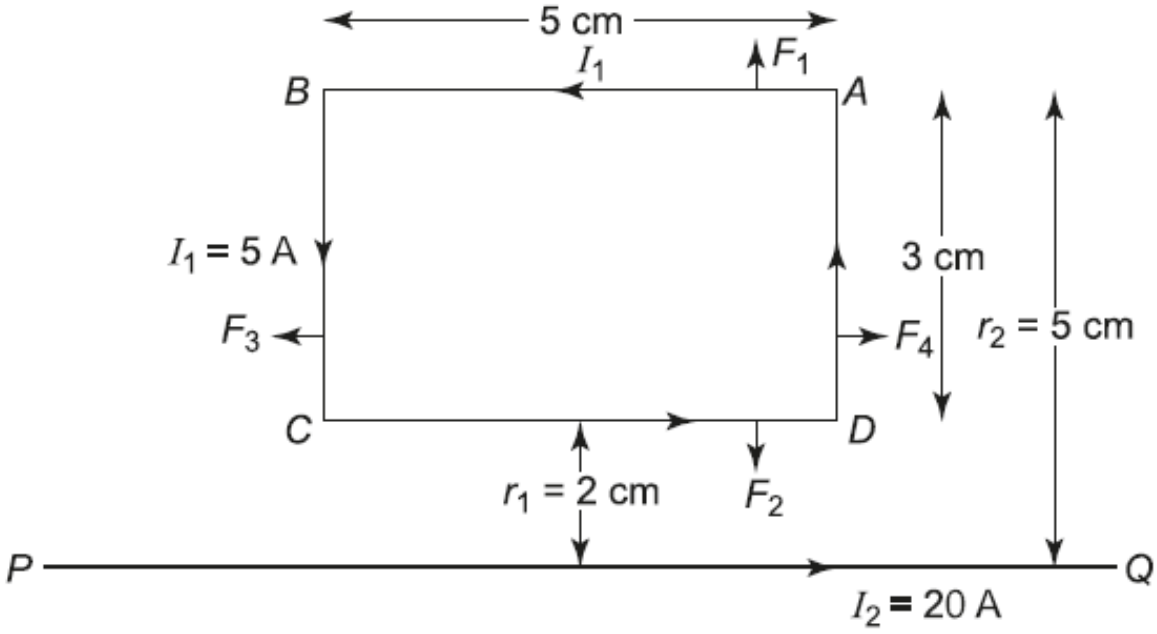
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A small rectangular loop $ABCD$ of sides $5\ \text{cm}$ and $3\ \text{cm}$ carries a current of $5\ \text{A}$. It is placed with its longer side parallel to a long straight conductor PQ of length $5\ \text{m}$ at a distance of $2\ \text{cm}$ from it as shown in Fig. If the current in PQ is $20\ \text{A}$, find the net force on the loop. Is the loop attracted towards PQ or repelled away from it?



(A) attracted

(B) repelled

(C) both

(D) none

Q.2

A charge q moves with a velocity v in a magnetic field $\mathrm{B} = \hat{\mathrm{i}} + 2\hat{\mathrm{j}} + 4\hat{\mathrm{k}}$. If the acceleration of the particle at time t is $\mathrm{a} = 2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - n\hat{\mathrm{k}}$, then the value of n is

(A) 1

(B) 2

(C) 3

(D) 4

Q.3

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A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B. The time taken by the particle to complete one revolution is

(A) $\frac{2\pi q^2 B}{m}$

(B) $\frac{2\pi qB}{m}$

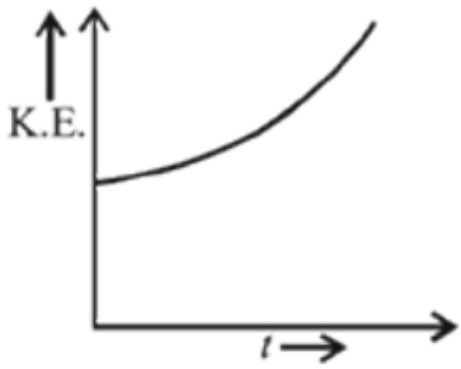
(C) $\frac{2\pi mq}{B}$

(D) $\frac{2\pi m}{qB}$

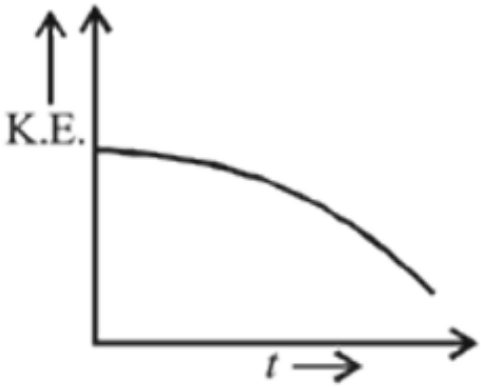
Q.4

A charged particle enters in a magnetic field in a direction perpendicular to the magnetic field. Which of the following graphs show the correct variation of kinetic energy of the particle with time t ?

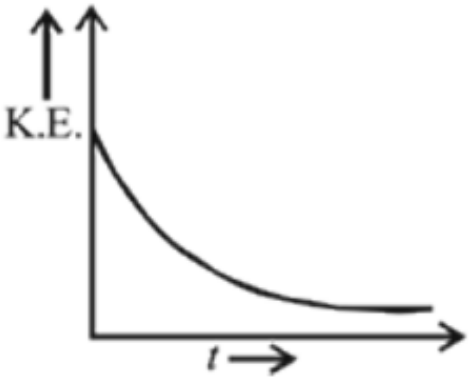
(A)



(B)

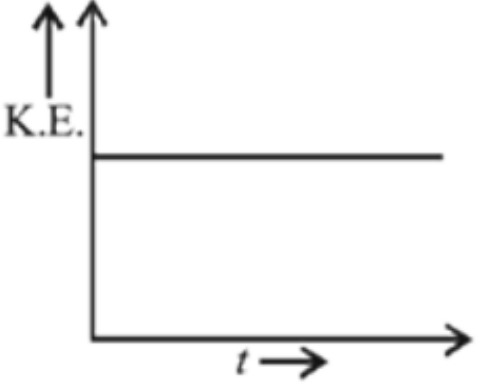


(C)



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(D)



Q.5

An electric charge $+q$ moves with velocity $\overrightarrow{\mathrm{v}} = 3\hat{i} + 4\hat{j} + \hat{k}$ in an electromagnetic field given by $\overrightarrow{\mathrm{E}} = 3\hat{\mathrm{i}} + \hat{\mathrm{j}} + 2\hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{B}} = \hat{i} + \hat{j} - 3\hat{\mathrm{k}}$ The y - component of the force experienced by $+q$ is :

(A) $11q$

(B) $5q$

(C) $3q$

(D) $2q$

Q.6

A beam of electrons is moving with constant velocity in a region having simultaneous perpendicular electric and magnetic fields of strength $20\mathrm{Vm}^{-1}$ and $0.5\ \mathrm{T}$ respectively at right angles to the direction of motion of the electrons. Then the velocity of electrons must be

(A) $8\ \mathrm{m/s}$

(B) $20\ \mathrm{m/s}$

(C) $40\ \mathrm{m/s}$

(D) $\frac{1}{40}\ \mathrm{m/s}$

Q.7

Two electron beams having their velocities in the ratio $1:2$ are subjected to identical magnetic fields acting at right angles to the direction of motion of electron beams. The ratio of deflection produced is:

(A) $2:1$

(B) $1:2$

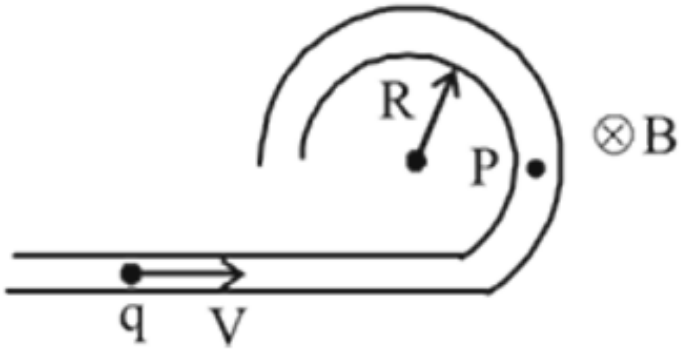
(C) $4:1$

(D) $1:4$

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Q.8

A charged particle moves insides a pipe which is bent as shown in fig. If $\mathrm{R} < \frac{\mathrm{mv}}{\mathrm{qB}}$, then force exerted by the pipe on charged particle at P is (Neglect gravity)



(A) toward center

(B) away from center

(C) zero

(D) none of these

Q.9

Under the influence of a uniform magnetic field a charged particle is moving in a circle of radius R with constant speed v. The time period of the motion

(A) depends on both R and v

(B) is independent of both R and v

(C) depends on R and not on v

(D) depends on v and not on R

Q.10

An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the $+\mathrm{x}$ -direction and a magnetic field along the $+z$ -direction, then

(A) positive ions deflect towards $+y$ -direction and negative ions towards $-y$ direction

(B) all ions deflect towards $+y$ -direction

(C) all ions deflect towards $-y$ -direction

(D) positive ions deflect towards $-\mathrm{y}$ -direction and negative ions towards $+y$ -direction.

Q.11

Three particles, an electron (e), a proton (p) and a helium atom (He) are moving in circular paths with constant

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speeds in the $x-y$ plane in a region where a uniform magnetic field B exists along z axis. The times taken by e, p and He inside the field to complete one revolution are $\mathrm{t_e}, \mathrm{t_p}$ and $\mathrm{t_{He}}$ respectively. Then,

(A) $\mathrm{t_{He} > t_p = t_e}$

(B) $\mathrm{t_{He} > t_p > t_e}$

(C) $\mathrm{t_{He} = t_p = t_e}$

(D) None of these

Q.12

A particle of specific charge $\frac{\mathrm{q}}{\mathrm{m}}=\pi \mathrm{Ckg}^{-1}$ is projected from the origin towards positive x -axis with a velocity of $10\ \mathrm{ms}^{-1}$ in a uniform magnetic field $\overrightarrow{\mathrm{B}}=-2 \hat{\mathrm{k}} \mathrm{T}$. The velocity $\overrightarrow{\mathrm{v}}$ of particle after time $t=1/12\ \mathrm{s}$ will be (in ms^{-1})

(A) $5[\hat{i}+\sqrt{3} \hat{j}]$

(B) $5[\sqrt{3} \hat{\mathrm{i}}+\hat{\mathrm{j}}]$

(C) $5[\sqrt{3} \hat{\mathrm{i}}-\hat{\mathrm{j}}]$

(D) $5[\hat{\mathrm{i}}+\hat{\mathrm{j}}]$

Q.13

A particle is projected in a plane perpendicular to a uniform magnetic field. The area bounded by the path described by the particle is proportional to

(A) the velocity

(B) the momentum

(C) the kinetic energy

(D) None of these

Q.14

There exist uniform magnetic and electric fields of magnitudes $1\ \mathrm{T}$ and $1\ \mathrm{V\ m}^{-1}$, respectively, along positive y-axis. A charged particle of mass $1 \mathrm{kg}$ and charge $1 \mathrm{C}$ is having velocity $1\ \mathrm{m\ s}$ along x -axis and is at origin at $\mathrm{t}=0$. Then, the coordinates of the particles at time π s will be

(A) $(0,1,2) \mathrm{m}$

(B) $\left(0,-\pi^2,-2\right)\ \mathrm{m}$

(C) $\left(2, \pi^2/2, 2\right)\ \mathrm{m}$

(D) $\left(0, \pi^2/2, 2\right)\ \mathrm{m}$

Q.15

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A particle of mass m and charge q moves with a constant velocity v along the positive x -direction. It enters a region containing a uniform magnetic field B directed along the negative z -direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is

(A) $\frac{qbB}{m}$

(B) $\frac{q(b-a)B}{m}$

(C) $\frac{qaB}{m}$

(D) $\frac{q(b+a)B}{2m}$

Q.16

Two particles X and Y having equal charges, after being accelerated through the same potential difference, enter a region of uniform magnetic field and describe circular paths of radii R_1 and R_2 respectively. The ratio of the mass of X to that of Y is

(A) $(R_1/R_2)^{1/2}$

(B) R_2/R_1

(C) $(R_1/R_2)^2$

(D) R_1/R_2.

Q.17

An electron, charge $-$e, mass m, enters a uniform magnetic field $\vec{B} = B\vec{i}$ with an initial velocity $\vec{v} = v_x\vec{i} + v_y\vec{j}$. What is the velocity of the electron after a time interval of t second?

(A) $v_x\hat{i} + v_y\hat{j} + \frac{e}{m}v_yBt\hat{k}$

(B) $v_x\hat{i} + v_y\hat{j} - \frac{e}{m}v_yBt\hat{k}$

(C) $v_x\hat{i} + \left(v_y + \frac{e}{m}v_yBt\right)\hat{j}$

(D) $v_x\hat{i} + \left(v_y + \frac{e}{m}v_yBt\right)\hat{i} + v_y\hat{j}$

Q.18

An alternating electric field, of frequency v is applied across the dees (radius $= R$) of a cyclotron that is being used to accelerate protons (mass m). The operating magnetic field (B) used in the cyclotron and the kinetic energy (K) of the proton beam, produced by it, are given by :

(A) $B = \frac{mv}{e}$ and $K = 2m\pi^2v^2R^2$

(B) $B = \frac{2\pi mv}{e}$ and $K = m^2\pi vR^2$

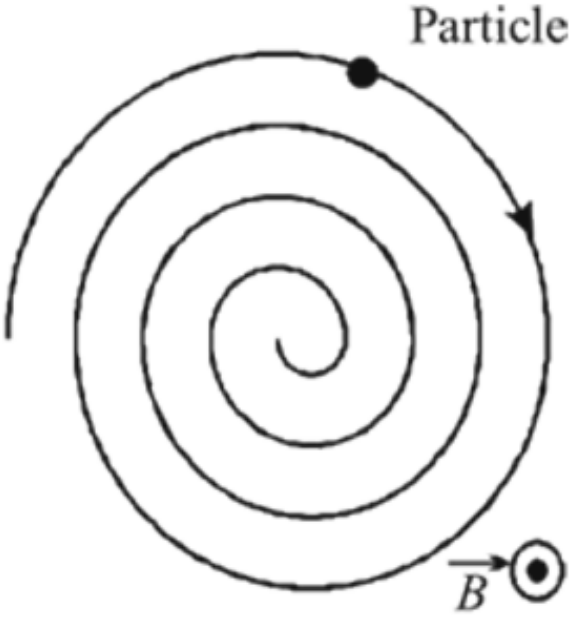
(C) $B = \frac{2\pi mv}{e}$ and $K = 2m\pi^2v^2R^2$

(D) $B = \frac{mv}{\rho}$ and $K = m^2\pi vR^2$

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Q.19

A uniform magnetic field is directed out of the page. A charged particle, moving in the plane of the page, follows a clockwise spiral of decreasing radius as shown.
A reasonable explanation is



(A) the charge is positive and slowing down

(B) the charge is negative and slowing down

(C) the charge is positive and speeding up

(D) the charge is negative and speeding up

Q.20

A particular region contains uniform electric and magnetic fields, $\vec{E} = E_0\hat{\imath}$ and $\vec{B} = B_0\hat{\imath}$ respectively. When a particle having mass and charge m and q respectively, enters the region with velocity $\vec{v} = v_0\hat{\jmath}$, it is observed that the speed of the particle becomes $2v_0$ after a time t. Find the value of t.

(A) $t = \frac{2mv_0}{qE}$

(B) $t = \frac{2Bq}{mv_0}$

(C) $t = \frac{\sqrt{3}Bq}{mv_0}$

(D) $t = \frac{\sqrt{3}mv_0}{qE}$

Answers & Solutions

Q.1 Answer:

attracted

Solution:

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Force exerted by PQ on AB is

$$F_1 = \frac{\mu_0 I_1 I_2}{2\pi r_1} \times AB$$
$$= \frac{\left(4\pi \times 10^{-7}\right) \times 5 \times 20 \times 5 \times 10^{-2}}{2\pi \times 5 \times 10^{-2}}$$
$$= 2 \times 10^{-5}\ \text{N}$$

(repulsive since I_1 and I_2 are in opposite directions)

Force exerted by PQ on CD is

$$F_2 = \frac{\mu_0 I_1 I_2}{2\pi r_2} \times CD$$
$$= \frac{\left(4\pi \times 10^{-7}\right) \times 5 \times 20 \times 5 \times 10^{-2}}{2\pi \times 2 \times 10^{-2}}$$
$$= 5 \times 10^{-5}\ \text{N}$$

(attractive since I_1 and I_2 are in the same direction)

From Fleming's L.H. rule, the magnetic field due to current in PQ is directed outwards (towards the reader) and perpendicular to the plane of the coil. Therefore, forces F_3 and F_4 on BC and AD are equal and opposite and hence cancel each other. Therefore, the net force on coil $ABCD$ is

$$F = F_2 - F_1 = 5 \times 10^{-5} - 2 \times 10^{-5} = 3 \times 10^{-5}\ \text{N}$$

(attractive). Hence coil is attracted towards PQ.

Q.2 Answer:

2

Solution:

$\mathrm{F} = q(v \times \mathrm{B})$. **Hence** F **is perpendicular to** B. **Therefore,**

$$\mathrm{F} \cdot \mathrm{B} = 0$$

or $$m\mathrm{a} \cdot \mathrm{B} = 0$$

$$\Rightarrow \mathrm{a} \cdot \mathrm{B} = 0 \quad (\because m \neq 0)$$

$$\Rightarrow (2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - n\hat{\mathrm{k}}) \circ (\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + 4\hat{\mathrm{k}}) = 0$$

$$\Rightarrow 2 + 6 - 4n = 0$$

$$\Rightarrow n = 2$$

So the correct choice is (b).

Q.3 Answer:

$\frac{2\pi \mathrm{m}}{\mathrm{qB}}$

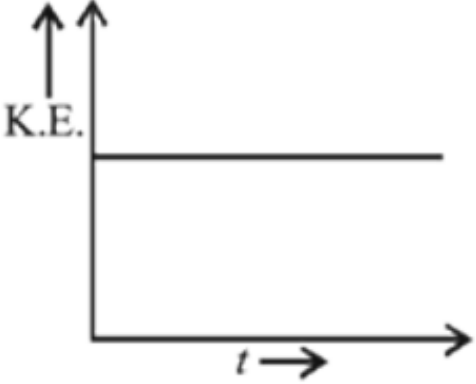
Solution:

Equating magnetic force to centripetal force, $\frac{mv^2}{r} = qvB \sin 90^\circ$

Time to complete one revolution, $\mathrm{T} = \frac{2\pi \mathrm{r}}{\mathrm{v}} = \frac{2\pi \mathrm{m}}{\mathrm{qB}}$

Q.4 Answer:

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Solution:

The change in K.E. is equal to work done by net force which is zero because the magnetic force is perpendicular to velocity.
K.E. remains constant.

Q.5 Answer:

$11q$

Solution:

Lorentz force acting on the particle

$$\vec{F} = \mathrm{q}[\overrightarrow{\mathrm{E}} + \overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{B}}]$$
$$= \mathrm{q}[3\hat{\mathrm{i}} + \hat{\mathrm{j}} + 2\hat{\mathrm{k}} + \hat{\mathrm{i}}(-12 - 1).$$
$$-\hat{j}(-9 - 1) + k(3 - 4)$$

Q.6 Answer:

$40\ \mathrm{m/s}$

Solution:

The electron moves with constant velocity without deflection. Hence, force due to magnetic field is equal and opposite to force due to electric field.

Q.7 Answer:

$2 : 1$

Solution:

Since, magnetic field is acting at right angles to the direction of motion of electron beams, so their paths will be circular.

$r = \frac{mv}{qB}$ **or** $r \propto v$

$\therefore \frac{r_1}{r_2} = \frac{v_1}{v_2} = \frac{1}{2}$

or $r_1 : r_2 = 1 : 2$

Now, as we know deflection of the electron beam $(\Delta d) \propto \frac{1}{\text{velocity}}$.

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So, ratio of deflection

$(\Delta d_1) : (\Delta d_2) = (v_2) : (v_1) = 2 : 1$

Q.8 Answer:

toward center

Solution:

Radius of circular path of the charged particles $\mathrm{r} = \frac{\mathrm{mV}}{\mathrm{qB}}$

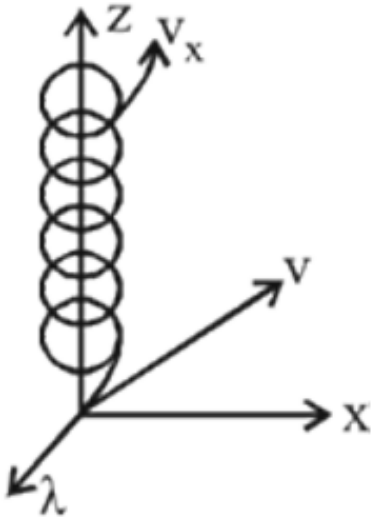
Here $\mathrm{R} < \mathrm{r}$**, hence the particle will press the outer wall of the pipe hence the force applied by the pipe on the particle should be towards the centre of the pipe.**

Q.9 Answer:

is independent of both R and v

Solution:

When a test charge q_0 **enters magnetic field** $\vec{B}$ **directed along** z **-axis, with a velocity** $\vec{v}$ **making angles** d **with the** z **-axis. The time period of the motion is independent of** R **and** v



Q.10 Answer:

all ions deflect towards $-y$ -direction

Solution:

The correct option is C All ions deflect towards - y direction. As the electric field is switched on, positive ion will start to move along positive x**-direction and negative ion along negative** x**-direction. Current associated with motion of both types of ions is along positive** x**-direction. According to Flemings left hand rule or right hand thumb rule, force on both types of ions will be along negative** y **-direction.**

Q.11 Answer:

$\mathrm{t_{He}} > \mathrm{t_p} > \mathrm{t_e}$

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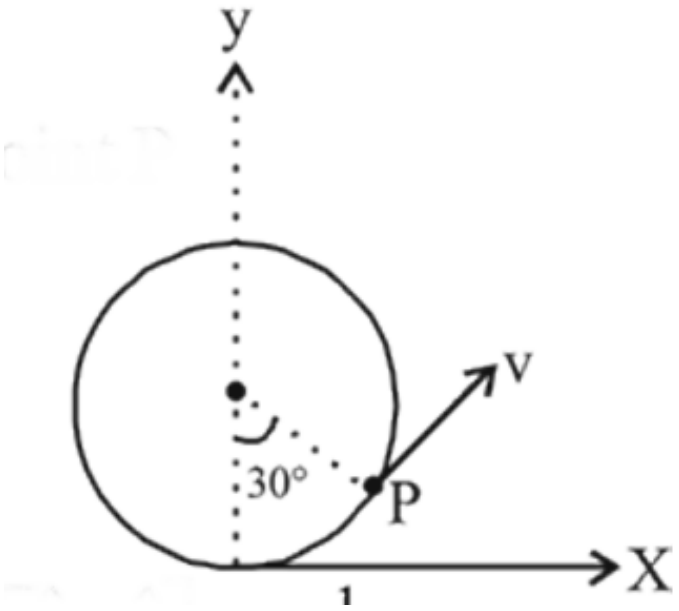
Solution:

$$T = 2\pi m/qB, \left(\frac{m}{q}\right)_{He} > \left(\frac{m}{q}\right)_{p} > \left(\frac{m}{q}\right)$$
$$\therefore \quad t_{He} > t_{p} > t_{e}$$

Q.12 Answer:

$5[\sqrt{3}\hat{i} + \hat{j}]$

Solution:



Time period $T = \frac{2\pi m}{qB} = \frac{2\pi}{\frac{q}{m}B} = \frac{2\pi}{\pi \times 2} = 1\ s$

Since the particle will be at point P **after time** $t = \frac{1}{12} = \frac{T}{12}$ **s, it is deviated by an angle** $\theta = \frac{2\pi}{12} = 30^{\circ}$

Hence, velocity at point P

$$\vec{v} = 10\left(\cos 30^{0}\hat{i} + \sin 30^{0}j\right)$$
$$= 10\left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1_{A}}{2}\right)$$
$$= 5\left(\sqrt{3}\hat{i} + 1_{j}\right)$$
$$= 5(\sqrt{3}\hat{i} + \hat{j})$$

Q.13 Answer:

the kinetic energy

Solution:

As $r = \frac{mv}{qB} = \frac{P}{qB}$

$\therefore$ **Area,** $A = \pi r^2 = \pi\left(\frac{P}{qB}\right)^2 = \frac{\pi P^2}{qB} = \frac{2m\pi}{qB}K$

Q.14 Answer:

$(0, \pi^2/2, 2)$ m

Solution:

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Since the particle will accelerate along the axis in which E is applied, it will have a helical path. The circular path is due to the magnetic field.

$\mathrm{T} = \frac{2\pi \mathrm{m}}{\mathrm{qB}} = 2\pi$

After π seconds, the particle will complete half the circle in the $z - x$ plane.

$\mathrm{r} = \frac{\mathrm{mv}}{\mathrm{qB}} = \frac{1\times1}{1\times1} = 1$

$\mathrm{X}-$ coordinate will be $\mathrm{x} = 0.$

$\mathrm{Z}-$ coordinate will be $\mathrm{z} = 2\mathrm{r} = 2$

Co-ordinate of the particle along $y - axis : y = ut + \frac{1}{2}at^2 = 0t + \frac{1}{2}\frac{qE}{m}t^2 = \frac{1}{2}\frac{\mathrm{qE}}{\mathrm{m}}\pi^2 = \frac{\pi^2}{2}$

Hence, the co-ordinate of the particle will be $\left(0, \frac{\pi^2}{2}, 2\right).$

Q.15 Answer:

$\frac{q(b-a)B}{m}$

Solution:

Radius of curvature has to be greater than the width of magnetic field for the particle to enter the region $\mathrm{x} > \mathrm{B}.$

$\therefore \frac{\mathrm{mv}}{\mathrm{qB}} > (\mathrm{b} - \mathrm{a})$

$\Rightarrow \mathrm{v} > \frac{\mathrm{qB(b-a)}}{\mathrm{m}}$

$\therefore \mathrm{v_{min}} = \frac{\mathrm{qB(b-a)}}{\mathrm{m}}$

Q.16 Answer:

$(R_1/R_2)^2$

Solution:

K.E. of first particle $= \frac{1}{2}m_1v_1^2 = qV$... (i)

K.E. of second particle $= \frac{1}{2}m_2v_2^2 = qV$...(ii)

After entering the magnetic field, a magnetic force acts on the charged particle which moves the charged particle in circular path of radius

$$R = \frac{\sqrt{2mK}}{qB}$$

Here , K, q, B are equal

$\therefore \quad R^2 \propto m \Rightarrow \frac{m_1}{m_2} = \frac{R_1^2}{R_2^2}$ From (i) and (ii)

Q.17 Answer:

$v_x\hat{i} + v_y\hat{j} + \frac{e}{m}v_yBt\hat{k}$

Solution:

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Force on a charged particle q of mass m moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ is given by $\vec{F} = q(\vec{v} \times \vec{B})$

Given: $\vec{v} = v_x\hat{i} + v_y\hat{j}$ **and** $\vec{B} = B\hat{i}$

$\therefore \vec{v} \times \vec{B} = -v_yB\hat{k}$

$\therefore \vec{F} = (-e)\left(-v_yB\hat{k}\right) = v_yBe\hat{k}[\because q = -e]$

Now, acceleration of electron, $\vec{a} = \frac{\vec{F}}{m} = \frac{v_yBe}{m}\hat{k}$ **. . ...(i)**

Velocity of an electron after time t seconds is $\vec{v}$**,**

$\therefore \vec{v} = \vec{v} + \vec{a}t$ **[where** $\vec{v}$ **is the initial velocity]**

$\Rightarrow \vec{v} = v_x\hat{i} + v_y\hat{j} + \left(\frac{v_yBe}{m}\right)t\hat{k}[$ **using (i)]**

Q.18 Answer:

$B = \frac{2\pi mv}{e}$ and $K = 2m\pi^2v^2R^2$

Solution:

Time period of cyclotron is $T = \frac{1}{v} = \frac{2\pi m}{eB}; B = \frac{2\pi m}{e}v; R = \frac{mv}{eB} = \frac{p}{eB}$

$\Rightarrow \quad P = eBR = e \times \frac{2\pi mv}{e}R = 2\pi m \cup R$

K.E. $= \frac{p^2}{2m} = \frac{(2\pi mvR)^2}{2m} = 2\pi^2mv^2R^2$

Q.19 Answer:

the charge is positive and slowing down

Solution:

$R = \frac{mv}{qB}; \vec{F} = q(\vec{v} \times \vec{B})$

As R **decreases,** v **will decrease. Clockwise motion for upward magnetic field** $\rightarrow$ **+ve**

Q.20 Answer:

$t = \frac{\sqrt{3}mv_0}{qE}$

Solution:

Given

$\vec{E} = E_0\hat{i}$ **and** $\vec{B} = B_0\hat{i}$

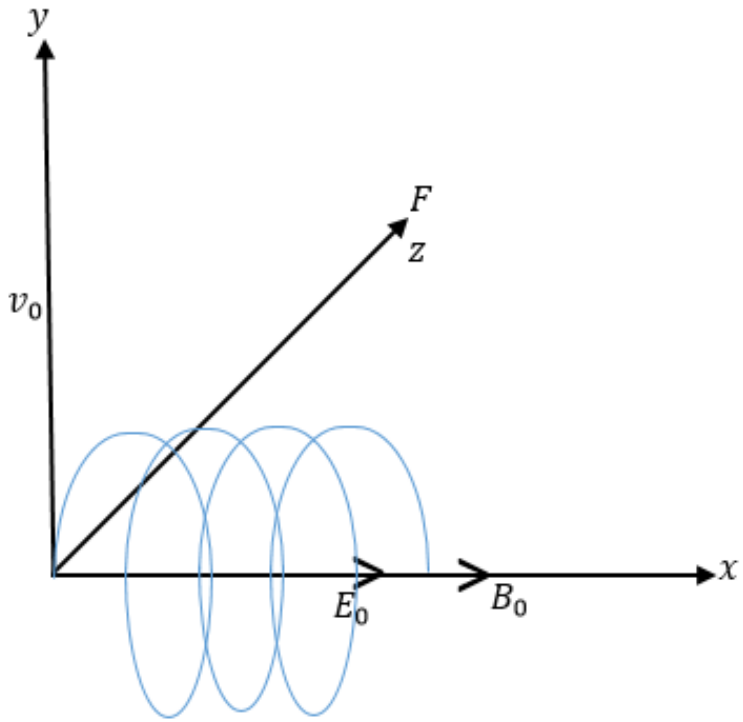
$\therefore \vec{E} \parallel \vec{B}$

$\vec{v} = v_0\hat{j}$

$\therefore \vec{v} \perp \vec{B}$

Hence the path of the particle is a helix with speed remaining constant for the circular path in the y-z plane.

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Magnitude of velocity of particle at any time t

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$

velocity along x axis will increase

$v_x = a_x t$

$\Rightarrow v = \sqrt{(a_x t)^2 + v_0^2}$ $\quad where, \quad v_y^2 + v_z^2 = v_0^2$

Given

$v = 2v_0$

$2{v_0}^2 = v_x^2 + v_0^2$

$v_x = \sqrt{3}v_0$

$a_x t = \sqrt{3}v_0$

$F = ma$ **and** $F_E = qE \Rightarrow qE = ma$

$\therefore a = \frac{qE}{m}$

$\frac{qE}{m}t = \sqrt{3}v_0$

$t = \frac{\sqrt{3}v_0 m}{qE}$

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